12TH BIOLOGY · CHAPTER 21 · FULL CHAPTER MCQS

Immunity

62 topic-wise MCQs from the complete chapter with answers and reasons

21.1 FIRST LINE OF DEFENCE

1. The first line of defence consists of skin and the:

(a) Digestive and respiratory tracts (b) Blood and lymph
(c) T cells and B cells (d) Liver and spleen

Answer: (a) Digestive and respiratory tracts — All are barriers at the body surface.

2. The epidermis of skin is how many cells thick?

(a) 2 to 5 (b) 10 to 30
(c) 50 to 100 (d) 100 to 200

Answer: (b) 10 to 30 — As stated in the textbook.

3. The outer layer of the epidermis is the:

(a) Dermis (b) Hypodermis
(c) Stratum corneum (d) Mucosa

Answer: (c) Stratum corneum — Its cells are continuously shed.

4. The protein that makes skin tough and impenetrable is:

(a) Collagen (b) Perforin
(c) Lysozyme (d) Keratin

Answer: (d) Keratin — It is formed by epidermal cells.

5. Oil and sweat glands lower the pH of the skin surface to:

(a) 1–2 (b) 3–5
(c) 7–8 (d) 9–10

Answer: (b) 3–5 — This acidity inhibits many microbes.

6. The enzyme in sweat and saliva that digests bacterial cell walls is:

(a) Amylase (b) Perforin
(c) Pepsin (d) Lysozyme

Answer: (d) Lysozyme — It attacks the cell walls of many bacteria.

7. Bacteria in swallowed food are mainly killed in the stomach by:

(a) Strong acids (b) Mucous
(c) Bile (d) Cilia

Answer: (a) Strong acids — Intestinal enzymes also kill bacteria.

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8. Airborne microbes in the bronchi are trapped by:

(a) Keratin (b) Lysozyme
(c) Sticky mucous (d) Acid

Answer: (c) Sticky mucous — Cilia sweep the mucous upward.

9. In an adult human, skin makes up what share of body weight?

(a) 5% (b) 40%
(c) 25% (d) 15%

Answer: (d) 15% — Skin is the largest organ.

21.2 SECOND LINE OF DEFENCE

10. The second line of defence is:

(a) Specific (b) Non-specific
(c) Acquired (d) Passive

Answer: (b) Non-specific — Like the first line.

11. Macrophages develop from:

(a) Neutrophils (b) Plasma cells
(c) B cells (d) Monocytes

Answer: (d) Monocytes — Monocytes are white blood cells.

12. A macrophage ingests a bacterium by:

(a) Exocytosis (b) Diffusion
(c) Endocytosis (d) Osmosis

Answer: (c) Endocytosis — Then lysozyme enzymes kill it.

13. Neutrophils use how many methods to kill pathogens?

(a) Three (b) One
(c) Two (d) Five

Answer: (a) Three — Phagocytosis, chemicals and DNA fibres.

14. Neutrophils trap bacteria with a network of:

(a) Protein fibres (b) Keratin
(c) Mucous (d) DNA fibres

Answer: (d) DNA fibres — The fibres bind and kill bacteria.

15. Natural killer cells attack:

(a) Virus-infected body cells (b) Free bacteria
(c) Antibodies (d) Red blood cells

Answer: (a) Virus-infected body cells — They also kill cancer cells.

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16. The proteins released by natural killer cells are:

(a) Interferons (b) Perforins
(c) Pyrogens (d) Histamines

Answer: (b) Perforins — They make a pore in the target membrane.

17. The surface marker that protects normal body cells is called:

(a) Antigen (b) Antibody
(c) Self-protein (d) Perforin

Answer: (c) Self-protein — Germ-killer cells do not attack cells having it.

18. Complement proteins are produced in the:

(a) Liver (b) Thymus
(c) Spleen (d) Bone marrow

Answer: (a) Liver — They circulate in an inactive state.

19. The membrane-attack complex is formed by the final:

(a) Two proteins (b) Five proteins
(c) Twelve proteins (d) Three proteins

Answer: (b) Five proteins — It punctures the microbe's membrane.

20. Interferons are released by:

(a) Bacteria (b) Mast cells
(c) Virus-infected cells (d) Plasma cells

Answer: (c) Virus-infected cells — They alert nearby cells.

21. Interferons belong to the class of proteins called:

(a) Antibodies (b) Pyrogens
(c) Perforins (d) Cytokines

Answer: (d) Cytokines — They work for communication between cells.

22. Inflammation is started by:

(a) Histamines and prostaglandins (b) Pyrogens
(c) Antibodies (d) Perforins

Answer: (a) Histamines and prostaglandins — They are released by infected or injured cells.

23. Which is NOT a sign of inflammation?

(a) Redness (b) Paleness
(c) Swelling (d) Warmth

Answer: (b) Paleness — Blood flow increases at the site.

24. Pyrogens are released by:

(a) B cells (b) Mast cells
(c) Macrophages (d) Red blood cells

Answer: (c) Macrophages — When they attack microbes.

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25. Pyrogens act on the:

(a) Thymus (b) Liver
(c) Cerebellum (d) Hypothalamus

Answer: (d) Hypothalamus — It is the body's thermostat.

26. Fever is also called:

(a) Anaphylaxis (b) Pyrexia
(c) Inflammation (d) Hypothermia

Answer: (b) Pyrexia — Temperature above 37 °C.

27. Body temperature above which value is often fatal?

(a) 100 °F (b) 101 °F
(c) 105 °F (d) 99 °F

Answer: (c) 105 °F — Above 103 °F is considered dangerous.

21.3 THIRD LINE OF DEFENCE

28. The third line of defence is:

(a) Specific (b) Non-specific
(c) Inborn (d) Mechanical

Answer: (a) Specific — It targets particular germs.

29. Immunity present at birth is:

(a) Acquired (b) Passive
(c) Innate (d) Artificial

Answer: (c) Innate — Also called inborn immunity.

30. Immunity in which ready-made antibodies are given is:

(a) Active (b) Natural active
(c) Innate (d) Passive

Answer: (d) Passive — It is short-lived.

31. Vaccination produces:

(a) Natural active immunity (b) Artificial active immunity
(c) Passive immunity (d) Innate immunity

Answer: (b) Artificial active immunity — Weakened antigens are administered.

32. The most effective form of immunity is:

(a) Natural active (b) Passive
(c) Artificial active (d) Innate

Answer: (a) Natural active — The body makes antibodies on natural exposure.

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33. T cells and B cells are together called:

(a) Monocytes (b) Phagocytes
(c) Lymphocytes (d) Mast cells

Answer: (c) Lymphocytes — They are white blood cells.

34. T cells mature in the:

(a) Bone marrow (b) Thymus
(c) Spleen (d) Liver

Answer: (b) Thymus — A small grey gland above the heart.

35. B cells mature in the:

(a) Thymus (b) Spleen
(c) Lymph nodes (d) Bone marrow

Answer: (d) Bone marrow — They do not travel to the thymus.

36. The T cells that initiate the immune response are:

(a) Helper (b) Cytotoxic
(c) Suppressor (d) Inducer

Answer: (a) Helper — They start both immune responses.

37. The T cells that terminate the immune response are:

(a) Helper (b) Inducer
(c) Suppressor (d) Cytotoxic

Answer: (c) Suppressor — They shut the response down.

38. The T cells that supervise T cell development in the thymus are:

(a) Helper (b) Suppressor
(c) Cytotoxic (d) Inducer

Answer: (d) Inducer — As listed in the textbook.

39. Antibodies are secreted by:

(a) Helper T cells (b) Plasma cells
(c) Macrophages (d) Mast cells

Answer: (b) Plasma cells — They are formed from B cells.

40. Antibody-mediated immunity is also called:

(a) Cellular immunity (b) Innate immunity
(c) Humoral immunity (d) Passive immunity

Answer: (c) Humoral immunity — Its key players are B cells.

41. T cell growth factor is produced by:

(a) Activated helper T cells (b) B cells
(c) Macrophages (d) Suppressor T cells

Answer: (a) Activated helper T cells — It starts the proliferation of T cells.

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42. Suppressor T cells shut down the cytotoxic response after:

(a) 1 to 2 hours (b) 1 to 2 years
(c) 1 to 2 days (d) 1 to 2 weeks

Answer: (d) 1 to 2 weeks — They multiply slowly.

43. Rapid response to a second infection is due to:

(a) Plasma cells (b) Memory cells
(c) Mast cells (d) Neutrophils

Answer: (b) Memory cells — They persist, probably for life.

44. Antibodies make up about what share of blood plasma protein?

(a) 2% (b) 80%
(c) 50% (d) 20%

Answer: (d) 20% — By weight.

45. An antibody molecule is made of how many polypeptide chains?

(a) Two (b) Three
(c) Four (d) Six

Answer: (c) Four — Two heavy and two light chains.

46. The chains of an antibody are held together by:

(a) Peptide bonds (b) Disulfide bridges
(c) Hydrogen bonds (d) Ionic bonds

Answer: (b) Disulfide bridges — The molecule is Y-shaped.

47. The antigen-binding site is formed by the:

(a) Variable regions (b) Constant regions
(c) Disulfide bridges (d) Stem of the Y

Answer: (a) Variable regions — Of one heavy and one light chain.

48. The small portion of an antigen that fits the binding site is the:

(a) Allergen (b) Epidermis
(c) Hybridoma (d) Determinant

Answer: (d) Determinant — Monoclonal antibodies bind only one of these.

49. Milstein and Kohler received the Nobel Prize in:

(a) 1974 (b) 1984
(c) 1994 (d) 1964

Answer: (b) 1984 — For producing monoclonal antibodies.

50. Hybridomas are formed from plasma cells and:

(a) Mast cells (b) T cells
(c) Myeloma cells (d) Stem cells

Answer: (c) Myeloma cells — Myelomas are cancer cells.

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51. Which protein was purified with monoclonal antibodies?

(a) Interferon (b) Insulin
(c) Keratin (d) Perforin

Answer: (a) Interferon — They purify specific molecules from mixtures.

52. Tumour necrosis factor (TNF):

(a) Causes fever (b) Makes antibodies
(c) Causes allergy (d) Kills tumour cells

Answer: (d) Kills tumour cells — TIL are being engineered to make it.

21.3.3 DISORDERS OF IMMUNE SYSTEM

53. Substances that cause allergies are called:

(a) Antibodies (b) Cytokines
(c) Pyrogens (d) Allergens

Answer: (d) Allergens — For example dust, pollen and cat dander.

54. Allergy involves which antibodies?

(a) G-antibodies (b) E-antibodies
(c) A-antibodies (d) M-antibodies

Answer: (b) E-antibodies — They bind to mast cells.

55. Histamines are released in allergy by:

(a) Plasma cells (b) Neutrophils
(c) Mast cells (d) Memory cells

Answer: (c) Mast cells — They lie around blood vessels.

56. Life-threatening swelling that makes breathing difficult is:

(a) Anaphylactic shock (b) Pyrexia
(c) Desensitization (d) Inflammation

Answer: (a) Anaphylactic shock — It is caused by histamines.

57. Injecting extracts of the antigens to treat allergy is called:

(a) Vaccination (b) Transplantation
(c) Desensitization (d) Gene therapy

Answer: (c) Desensitization — It produces G-antibodies.

58. In myasthenia gravis, antibodies attack receptors of:

(a) Insulin (b) Acetylcholine
(c) Histamine (d) Adrenaline

Answer: (b) Acetylcholine — On skeletal muscle cells.

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59. In rheumatoid arthritis, antibodies damage:

(a) Skeletal muscles (b) Skin
(c) Pancreas (d) Cartilage of joints

Answer: (d) Cartilage of joints — It is an autoimmune disease.

60. Type I diabetes results from destruction of:

(a) Insulin-producing cells (b) Liver cells
(c) Mast cells (d) Muscle receptors

Answer: (a) Insulin-producing cells — In the pancreas.

61. The best transplant donor is a/an:

(a) Parent (b) Identical twin
(c) Sibling (d) Unrelated donor

Answer: (b) Identical twin — The self-proteins match 100%.

62. Transplant rejection is reduced by drugs that:

(a) Strengthen immunity (b) Raise temperature
(c) Suppress immunity (d) Release histamine

Answer: (c) Suppress immunity — They also reduce the ability to fight infections.

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